Deterministic Optimization

Geometric Aspects of Linear Optimization

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Representation of Polyhedron

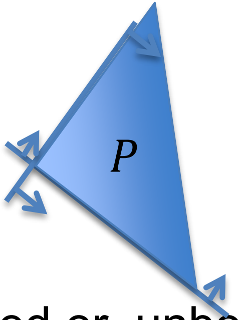
Geometric Aspects of Linear Optimization

Learning Objectives for this lesson

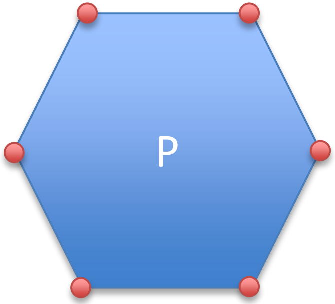
- Recognize a fundamental geometric characterization of a polyhedron:
  - A polyhedron can be "generated" using a finite number of extreme points and extreme rays

Polyhedron Representations

- Halfspace representation: A **polyhedron** is the **intersection** of **a finite number** of **halfspaces.**
  - $$P = \{\boldsymbol{x} \in R^n \mid \boldsymbol{a}_1^\top \boldsymbol{x} \le b_1, \; \ldots, \; \boldsymbol{a}_m^\top \boldsymbol{x} \le b_m\}$$

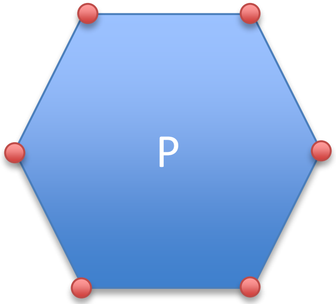


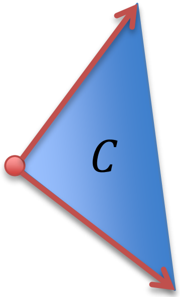
  - This representation applies to all polyhedra, bounded or unbounded
- Extreme-point representation: A **bounded** polyhedron is the convex hull of all its extreme points.
  - Convex hull can only generate a bounded set.
  - How to extend extreme-point rep. to an **unbounded** polyhedron?



Convex Hull & Conic Hull

- Convex combination of extreme points and convex hull
- Conic combination of extreme rays and conic hull





- How to the two together to describe a general polyhedron?

Example

- $P = \{(x_1, x_2): x_1 + x_2 \geq 1, x_1 - x_2 \leq 2, x_1, x_2 \geq 0\}$
- Extreme points:

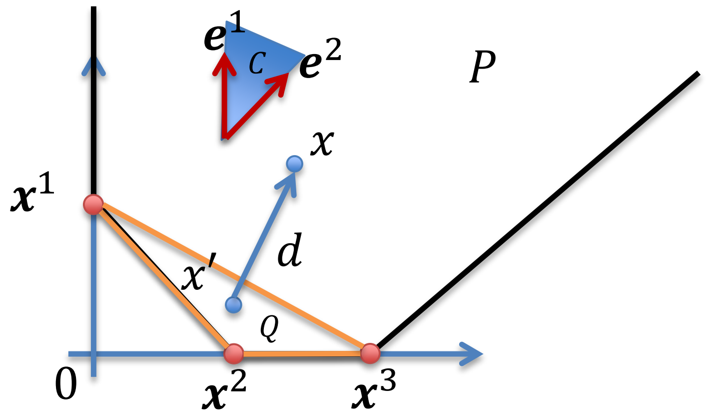
$$\boldsymbol{x}^1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \boldsymbol{x}^2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \boldsymbol{x}^3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

- Convex hull of $\boldsymbol{x}^1, \boldsymbol{x}^2, \boldsymbol{x}^3$: Q
- Extreme rays:

$$\boldsymbol{e}^1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \boldsymbol{e}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Conic hull of $\boldsymbol{e}^1$ and $\boldsymbol{e}^2$: C
- Any point $x \in P$ can be written as

$x = x' + d$, where $x' \in \text{conv}\{\boldsymbol{x}^1, \boldsymbol{x}^2, \boldsymbol{x}^3\}$ and $d \in \text{conic}\{\boldsymbol{e}^1, \boldsymbol{e}^2\}$



Example cont.

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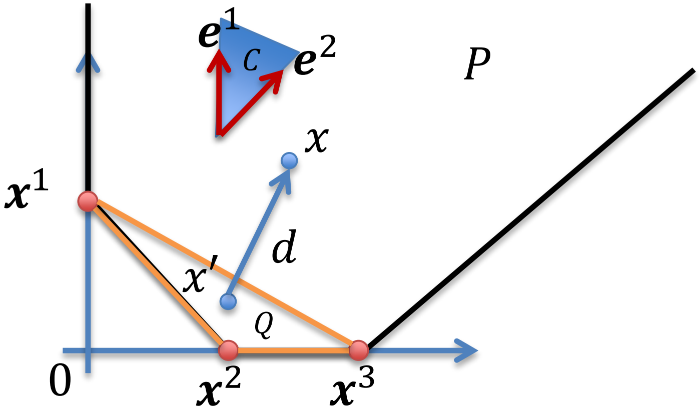

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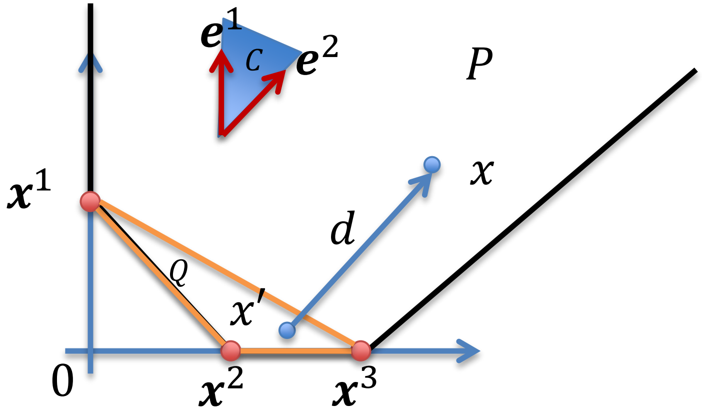

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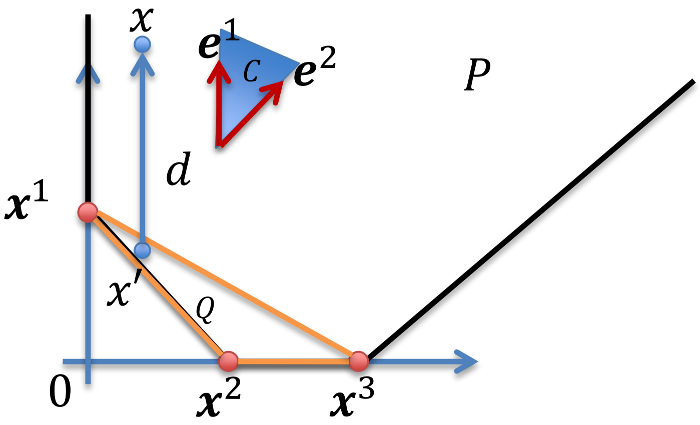

Weyl-Caratheodory Theorem

- Any point x in a polyhedron P that has at least one extreme point can be written as a sum of two vectors,

$$x = x' + d$$

where x' is in the convex hull of its extreme points and d is in the conic hull of its extreme rays.

    - If P does not have an extreme ray, then $d = 0$.

Theorem [Weyl-Caratheodory] Any nonempty polyhedron P with at least one extreme point can be formed by its extreme points $\boldsymbol{x}^1, \ldots, \boldsymbol{x}^m$ and its extreme rays $\boldsymbol{e}^1, \ldots, \boldsymbol{e}^k$ as follows

$$P = \mathrm{conv}\{\boldsymbol{x}^1, \ldots, \boldsymbol{x}^m\} + \mathrm{conic}\{\boldsymbol{e}^1, \ldots, \boldsymbol{e}^k\}$$

GTx

Summary

- We learned:
  - How to decompose a polyhedron into convex hull of its extreme points (if any) and conic hull of its extreme rays (if any)
  - Weyl-Caratheodory Theorem is a fundamental characterization of polyhedron
  - Polyhedron is a finitary structure, i.e. it is characterized by a finite number of extreme points and extreme rays
  - Extreme points and rays will be very important for algorithms